RES-5 2PI common-mode monotonicity: the R-U10-3 mechanism extends from the

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The ordered-BCC parallel reduced the screening route to necessary-but-not-sufficient and named the load-bearing residual: the NON-PERTURBATIVE higher-loop common-mode cancellation. The operator directed the structural target – extend the R-U10-3 one-loop mechanism to the 2PI effective action. This note carries out the four steps and finds that the common-mode MECHANISM extends; the residual is the a0-bound on the pattern-dependent skeleton. Scope: the structural extension; the rigorous a0-bound is the residual.

2. The 2PI effective action

In the Cornwall–Jackiw–Tomboulis formalism, the free energy is the stationary value of

$$\Gamma[G;\mathcal{P}] = \tfrac{1}{2}\,\mathrm{Tr}\ln G^{-1} + \tfrac{1}{2}\,\mathrm{Tr}\big[(K_0 + \lambda'\mathcal{P}^2)\,G\big] + \Gamma_2[G],$$

where $\Gamma_2[G]$ is the sum of two-particle-irreducible SKELETON diagrams and $F[\mathcal{P}] = \Gamma[G_*(\mathcal{P});\mathcal{P}]$ at the stationary $\delta\Gamma/\delta G = 0$.

3. Step 1 – Γ_2 is a pattern-independent functional

$\Gamma_2[G]$ depends on G and the bare vertices ONLY; the condensate $\mathcal{P}$ enters Γ only through the tree, the explicit $\lambda'\mathcal{P}^2 G$ term, and implicitly through $G_*(\mathcal{P})$. This is the structural fact that makes the common-mode split possible: the strong skeleton content is the SAME functional for $\mathcal{P}$ and $\mathcal{R}_H$.

4. Step 2 – Feynman–Hellmann and the common-mode split

At the stationary point, $\delta\Gamma/\delta G|_{G_*} = 0$, so the implicit $dG_*/d\mathcal{P}$ term vanishes and

$$\frac{dF}{d\mathcal{P}} = \frac{\partial\Gamma}{\partial\mathcal{P}}\bigg|_{G_*} \quad \text{(explicit only).}$$

The strong-fluctuation content sits in G_*, but the $\mathcal{P}$-derivative sees only the explicit condensate coupling. Writing $G_*(\mathcal{P}) = G_*^{\mathrm{common}}(I) + \delta G_*^{\mathrm{pd}}(\mathcal{P})$ (the common dressed sea plus the pattern-dependent variation), the free energy splits as

$$F[\mathcal{P}] = \underbrace{\Gamma[G_*^{\mathrm{common}};\,\langle\cdot\rangle]}_{\text{common, cancels}} + \underbrace{\Delta\Gamma^{\mathrm{pd}}[\mathcal{P}]}_{O(a_0)}.$$

5. Step 3 – common-mode cancellation at the dressed G_*

At fixed intensity I the common dressing $D_0(I) = \hat{r} = r_R + 2\lambda' I$ is pattern-independent, so $\Gamma[G_*^{\text{common}}; \cdot]$ – including the strong $\Gamma_2[G_*^{\text{common}}]$ – is the SAME for $\mathcal{P}$ and $\mathcal{R}_H$ and CANCELS in the difference. This is the 2PI extension of the R-U10-3 dressing-swap: the strong common sea cancels at the self-consistent level, not just at one loop.

6. Step 4 – the residual and the operator-monotone sign

The surviving difference is the pattern-dependent part. Its LEADING piece is the dressed log-det $\frac{1}{2}\mathrm{Tr}[\ln(G_*^{-1}(\mathcal{P})) - \ln(G_*^{-1}(\mathcal{R}_H))]$, which retains the R-U10-3 operator-monotone sign because $\lambda' = 3u_{\text{eff}} > 0$ is repulsive (the dressed condensate self-energy is PSD). The remaining piece is the pattern-dependent SKELETON variation

$$\boxed{\Delta\Gamma_2^{\text{pd}} = \Gamma_2[G_*(\mathcal{P})] - \Gamma_2[G_*(\mathcal{R}_H)] = O(a_0)\,, \quad a_0 = 0.096,}$$

since $G_*(\mathcal{P})$ and $G_*(\mathcal{R}_H)$ differ only by the a_0-suppressed δG_*^{pd}. The LOAD-BEARING RESIDUAL is the rigorous bound $|\Delta\Gamma_2^{\text{pd}}| \le c\, a_0\, \Delta F_{\text{margin}}$ with c controlled – obtained from the common-mode structure (non-perturbatively), NOT a skeleton-by-skeleton perturbative estimate.

7. Status

T3 PROOF SKETCH. The common-mode MECHANISM extends to the 2PI effective action (pattern-independent Γ_2 + Feynman–Hellmann + common dressing + repulsive monotone sign). The leading dressed selection retains ≥ 0; the residual is the a_0-bound on the pattern-dependent skeleton. **OPEN:** the rigorous $|\Delta\Gamma_2^{\text{pd}}| \le c\, a_0\, \Delta F_{\text{margin}}$ (the constant c from the δG_*^{pd} sensitivity of Γ_2). No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. **B1 closure** $\Longleftrightarrow$ this a0-skeleton bound, for which the common-mode mechanism is now established at 2PI.

8. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"$\delta G_*^{\text{pd}} = O(a_0)$ is assumed, not proved."* VALID-with-mitigation: G_* solves the gap equation with self-energy $\lambda'\mathcal{P}^2 + \Sigma_2[G_*]$; the pattern-specific part of the source is $\lambda'(\mathcal{P}^2 - \langle\mathcal{P}^2\rangle) = O(a_0)\hat{r}$, so $\delta G_*^{\text{pd}} = O(a_0)$ to leading order in the gap-equation linearisation. The rigorous bound (including the self-consistent feedback) is the residual.

(β) *"Feynman–Hellmann needs the bare, attractive coupling."* VALID-and-handled: $dF/d\mathcal{P}$ uses the bare action, but the SIGN of the fluctuation determinant is governed by the DRESSED $\lambda' > 0$ (the self-consistent $\hat{r}$ absorbs the attractive bare u); the tree/sextic and the dressed log-det together give the R-U10-3 ≥ 0, which this note extends – it does not re-derive the leading sign.

(γ) *"Does this close RES-5?"* DISMISSED: it extends the MECHANISM and sharpens the residual to the a_0-skeleton bound; it does not prove that bound. B1 stays T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *structure* – $\Gamma_2[G]$ pattern- independent (CJT fact); *common dressing* – $\hat{r}(I) = 0.337$ pattern- independent; *sign* – $\lambda' = 8.06 > 0$ repulsive (monotone ≥ 0); *residual scale* – $\Delta\Gamma_2^{\text{pd}} = O(a_0) = O(0.096)$; *logic* – common sea cancels by stationarity, residual a_0-suppressed.

9. Result footer

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Result ID:                 B1-RH-ENUM / RES-5 2PI common-mode monotonicity extension
Precise statement:         In the CJT 2PI action Gamma[G;P] = (1/2)Tr ln G^-1 +
                           (1/2)Tr[(K0+lam'P^2)G] + Gamma_2[G], (i) Gamma_2 is a
                           pattern-independent functional; (ii) stationarity =>
                           dF/dP = dGamma/dP|_{G_*} (Feynman-Hellmann); (iii) the
                           common dressing D_0(I) is pattern-independent so the
                           common sea (incl. Gamma_2[G_*^common]) cancels in F[P]-
                           F[R_H]; (iv) lam'>0 repulsive gives the R-U10-3 operator-
                           monotone leading sign. The common-mode MECHANISM extends
                           to 2PI; the residual is the a0-bound |Gamma_2[G_*(P)]-
                           Gamma_2[G_*(R_H)]| = O(a0) <= c a0 Delta F_margin.
Scope:                     Structural extension; anchor mu^2. The rigorous a0-bound
                           (constant c) is OPEN.
Dependencies:              res5-orderedbcc-parallel (the residual); neargap-common-
                           mode-resolution (R-U10-3, one-loop); res5-commonmode-
                           envelope (a0); Math424-AddA.
Evidence grade:            ANALYTIC (2PI structure) + EXECUTED
                           (res5_2pi_commonmode_monotonicity.py 5/5).
Reproduction command:      python codes/vacuum/res5_2pi_commonmode_monotonicity.py
Expected output:           r_hat=0.337 common; lam'=8.06>0; a0=0.096; 5/5 PASS.
Falsification gate:        A pattern-dependent part of Gamma_2 that is O(1) not O(a0)
                           (would break the common-mode extension); or delta G_*^pd
                           not O(a0) (self-consistent feedback amplifying the pattern
                           dependence).
Tier before / after:       No tier flip. B1 T6 on {H-LAYER}. RES-5 residual sharpened
                           to the a0-bound on the pattern-dependent 2PI skeleton; the
                           common-mode mechanism is established at 2PI.
No-overclaim statement:    This does NOT close RES-5. It extends the common-mode
                           MECHANISM to 2PI (structural) and sharpens the residual to
                           |Delta Gamma_2^pd| <= c a0 Delta F_margin; the constant c
                           (the self-consistent skeleton sensitivity) is the residual.
                           B1 stays T6 on {H-LAYER}.
Next required action:      Bound c: linearise the 2PI gap equation to get
                           delta G_*^pd = O(a0) rigorously, then bound the Gamma_2
                           sensitivity dGamma_2/dG at G_* (the screened susceptibility
                           from res5-2pi-resummation-strategy enters here).
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